

COMP5411 Course Project Guidelines

Fall 2026, HKUST

General Information

All assignments in this course are related to the course project. The project is open-ended and can be done in groups of 2.

Pick an interesting project, effect, or idea and discuss it with the instructor before committing to it. You are encouraged to be creative, but your project must remain firmly grounded in computer graphics.

Technical Requirements

Your implementation must include strong geometry processing *and* rendering components using advanced techniques (for example, GPU shaders and related methods).

You may use existing libraries for mesh processing and standard graphics APIs for rendering. Using frameworks such as `three.js` with WebGL is encouraged, but you should *not* use systems that automate most rendering aspects of the project (for example, full game engines), since this is a graphics programming class.

You may also use AI tools to help you in your project. In the final report, you must clearly and specifically document which parts of the code you implemented yourself and which parts were generated by AI tools or taken from external libraries.

Project Ideas

The following examples are intended to give you a sense of appropriate project scope and direction:

- An innovative particle animation system with interesting particle generation, simulation, and rendering.
- A creative non-photorealistic mesh rendering method (for example, stylized hatching, painterly shading, or outline-based effects).
- A new mesh smoothing or filtering operator with an accompanying visualization and analysis of its behavior.

It is acceptable to extend an existing project or to work on something related to your research. In that case, you must clearly disclose what is new in this project compared to

prior work.

Scope and Topic Approval

If you would like to pursue a topic that is only tangentially related to geometry or rendering, you *must* discuss it with the instructor **before** sending a formal proposal. The main objective is to ensure that your project remains suitable for a graduate-level graphics programming course.

More details on possible topics, milestones, and the project process will be discussed in class and posted on the course website.